

The cooked food derived carcinogen 2-amino-1-methyl-6-phenylimidazo[4,5-b] pyridine is a potent oestrogen: a mechanistic basis for its tissue-speci...



The cooked meat carcinogen 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) induces tumours of the breast, colon and prostate in rats. Here we show that in addition to its well-established genotoxicity, which can be detected at concentrations $>10(-6)$ M, PhIP is also oestrogenic. In COS-1 cells transiently transfected with an oestrogen-responsive reporter gene, PhIP ($10(-10)$ - $10(-6)$ M) mediated transcription through oestrogen receptor (ER) alpha, but not ER-beta, and inhibition by the pure ER antagonist ICI 182 780 demonstrated a requirement for a functional ER. In contrast, the structurally related food-derived carcinogen 2-amino-3,8-dimethylimidazo[4,5-f]quinoxaline (MeIQx) failed to induce reporter gene transcription. Additionally, we show that in a hormonally responsive breast cancer cell line (MCF-7 cells), PhIP induced transcriptional activation using endogenously expressed ER. Examination of the genotoxic potential of PhIP using a model mammalian cell mutation assay (hprt(-) locus) demonstrated that the genetic toxicology of PhIP was readily detectable, but separate, in terms of effective concentration, from its oestrogenic activity. To determine whether the oestrogenicity of PhIP could mediate oestrogen-dependent responses such as cell growth, we examined the growth of hormonally responsive cells (MCF-7 cells). We show that PhIP can stimulate cell proliferation and, again, this was dependent upon a functional ER. Using ligand blotting, we further show that PhIP can stimulate the expression of progesterone receptor (PR-A and PR-B) and c-MYC and activate the MAPK signal transduction pathway. These responses were similar to that produced by oestradiol, in terms of temporal aspects, potency and a requirement for a functional ER. Each of these dose-dependent mitogenic responses occurred at concentrations of PhIP (approximately $10(-9)$ - $10(-11)$ M) that are likely to be equivalent to systemic human exposure via consumption of cooked meat. Thus PhIP can induce cellular responses that encompass altered gene expression and mitogenesis. We suggest that the combination of genetic toxicology and oestrogen-like promotion of genomic and cellular events provide a mechanism for the tissue-specific tumorigenicity of this compound.